

10	(a) pulse (of ultrasound) B1 produced by quartz / piezo-electric crystal (1) reflected from boundaries (between media) B1 reflected pulse detected B1 by the ultrasound transmitter (1) signal processed and displayed B1 intensity of reflected pulse gives information about the boundary (1) time delay gives information about depth (1) <i>(four B marks plus any two from the four, max. 6)</i> B2 [6]	
	(b) shorter wavelength B1 smaller structures resolved / detected <i>(not more sharpness)</i> B1 [2]	
	(c) (i) $I = I_0 e^{-\mu x}$ C1 ratio = $\exp(-23 \times 6.4 \times 10^{-2})$ C1 = 0.23 A1 [3]	
	(ii) later signal has passed through greater thickness of medium M1 so has greater attenuation / greater absorption / smaller intensity A1 [2]	

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